

Clinical Category Linker: Strategy and Purpose

Objective

The **Category Linker** is the "Reduce" stage of our Map-Reduce pipeline. It is responsible for taking independent clinical extractions from 15 separate ADA guideline JSON files and consolidating them into five domain-isolated datasets optimized for Neo4j Knowledge Graph ingestion.

Why Domain-Isolated Databases?

Instead of a single "Global" Knowledge Graph, we are creating five category-specific databases (**Assessment, Education, Pharma, Complications, Special Populations**). This provides three critical benefits:

1. **Contextual Integrity:** Clinical targets often vary by context (e.g., A1C targets in Pregnancy vs. General Adults). Isolation prevents the graph from blending contradictory advice.
2. **Reduced Retrieval Noise:** By routing queries to specific sub-graphs, we eliminate irrelevant nodes, improving the accuracy of GraphRAG responses.
3. **Scoped Deduplication:** Entity resolution (merging "A1C" and "HbA1c") occurs within the category, ensuring that relationships are linked to a single, canonical node within that domain.

Key Features of the Logic

- **Normalization (MD5 Hashing):** Node IDs are generated by hashing the normalized clinical name prefixed by the category. This ensures that the same medication or test across different files in one category links to the same Neo4j node.
- **Traceability (Explainability):** The linker "stacks" metadata. A single node in the CSV maintains an array of all original sources and page numbers. It also generates a `traceability.json` for manual audit.
- **Neo4j Compliance:** Outputs use strict batch-import headers (`:ID` , `:LABEL` , `string[]`) to enable immediate ingestion via `LOAD CSV` or the Neo4j Data Importer.

Current State

- **Input:** 15 `_entities.json` files (including Chapter 4).
- **Processing:** Semantic normalization using a medical synonym map.
- **Output:** 5 sub-folders, each containing `nodes.csv` , `relationships.csv` , and a `traceability.json` file.